

MedLedger Health Exchange - Narrative

MedLedger Clinical Exchange is the hospital group's shared integration backbone for patient portal access, insurer eligibility, imaging, laboratory distribution, and emergency break-glass access. The edge is internet-facing for patients and payer integrations, while clinical systems and archives remain in the restricted clinical data zone. The architecture has become difficult because vendor diagnostics, emergency access, and modern AI imaging pilots now touch the same trust fabric.

Patients authenticate through the Identity Gateway and request records, appointments, and lab results through the Clinical Integration Bus. The bus orchestrates chart reads from the Electronic Health Record Store, retrieves imaging from the PACS Archive, and brokers insurer API traffic for authorizations and claims status. Break-glass access is provided by a dedicated Break-Glass Service with strong audit requirements, yet the service can reach clinical repositories when staff are under duress or during emergency care.

The Vendor Support Bastion is the only approved path for remote vendor access. Support staff sometimes need temporary privileged sessions to imaging or lab infrastructure. The organization is anxious about ransomware propagation from vendor access, insider misuse of emergency overrides, and inadvertent disclosure of protected health information in support notes, AI outputs, or insurer callbacks.